De La Salle Holy Cross College

CLASS TEST

17 June 2025

Grade 10

MATHEMATICS

Functions

Examiner:	Mrs Moodley	**Time:**	45 min
Moderators:	Mr Matthyser, Mrs De Freitas and Mr Guilhermino	**Marks:**	45

Name:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 8 pages and 3 questions. Extra space is on page 8.
2. Please check that your question paper is complete.
3. Answer the questions in the space provided on this question paper.
4. A non-graphical, non-programmable calculator may be used. Ensure that it is set in DEGREE mode.
5. ALL necessary calculations must be shown clearly. Answers only will not necessarily be awarded full marks.
6. Diagrams and graphs are not necessarily drawn to scale.
7. Round off final answers to ONE decimal place, unless stated otherwise.
8. It is in your own interest to write legibly and to present your work neatly.

	Q 1	Q 2	Q 3	Total
Mark				
Total	7	18	20	45

QUESTION 1

Given $f(x) = 5 - 3x$.

(a) Evaluate the following:

(1) $f(-1)$ (1)

(2) $3 \cdot f(k)$ (2)

(3) $f(f(0))$ (2)

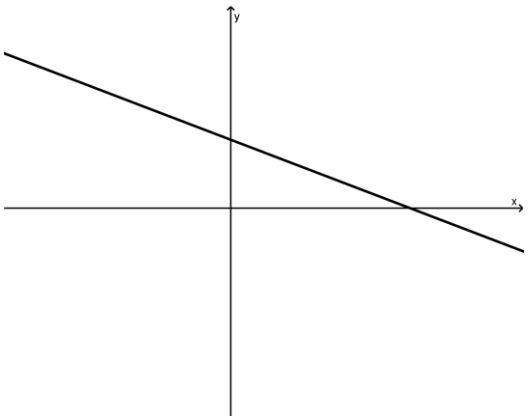
(b) Describe in words how f was transformed to become $y = -3 - 3x$. (2)

[7]

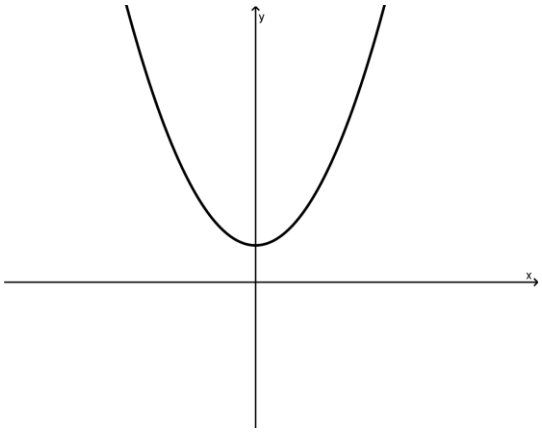
QUESTION 2

(a) Match each graph to the correct description of its equation.
Options may be used once only. (3)

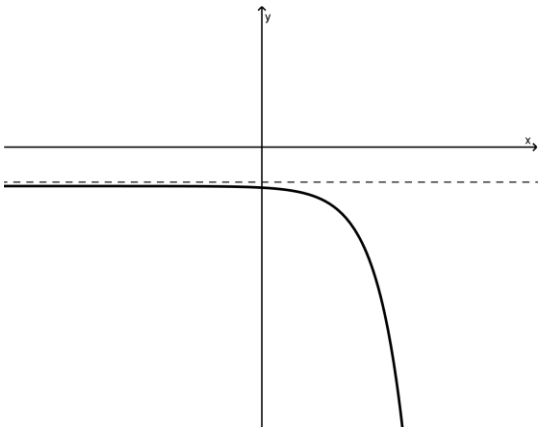
Graph 1: $y = ax + q$



Graph 2: $y = ax^2 + q$



Graph 3: $y = a \cdot b^x + q$



Options	
A:	$a>0$ and $q>0$
B:	$a<0$ and $q<0$
C:	$a>0$ and $q<0$
D:	$a<0$ and $q>0$

ANSWERS: **Option**

Graph 1: _____

Graph 2: _____

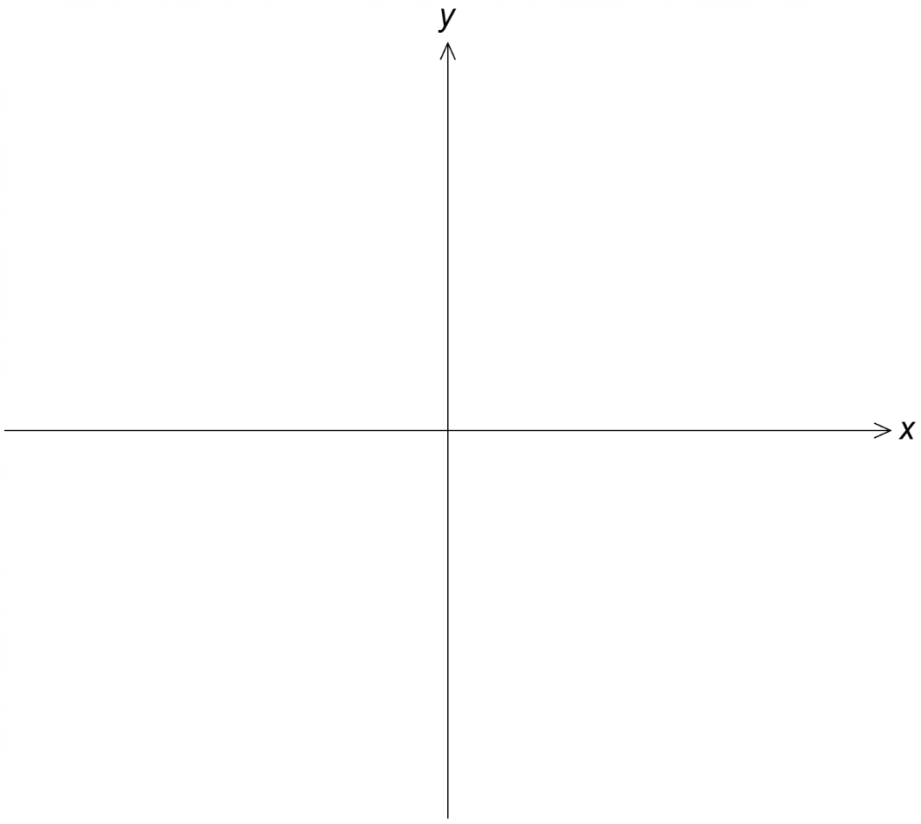
Graph 3: _____

(b) Consider $g(x)=\frac{9}{x}-3$.

(1) Determine the coordinates of the x-intercept(s) of g. (2)

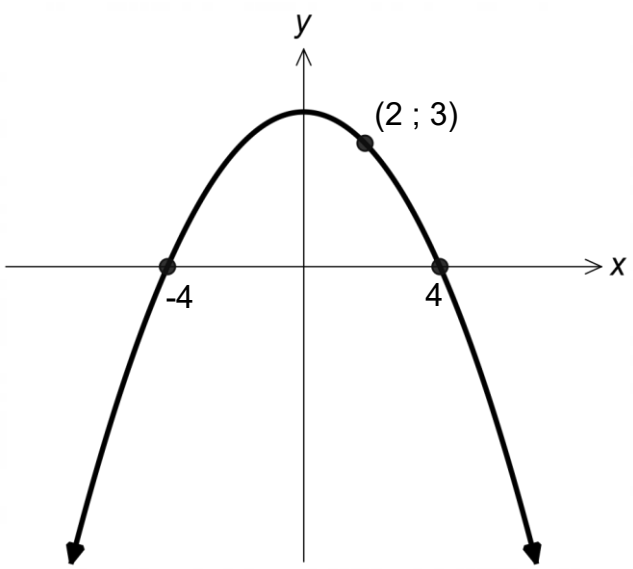
(2) Write down the equations of the asymptotes of g. (2)

(4) Hence, or otherwise, make a neat sketch graph of g.
Label all pertinent information.
Indicate and label one miscellaneous point on your graph. (5)



(5) One of the axes of symmetry of g has no points of intersection with g. Write down the equation of this axis of symmetry. (2)

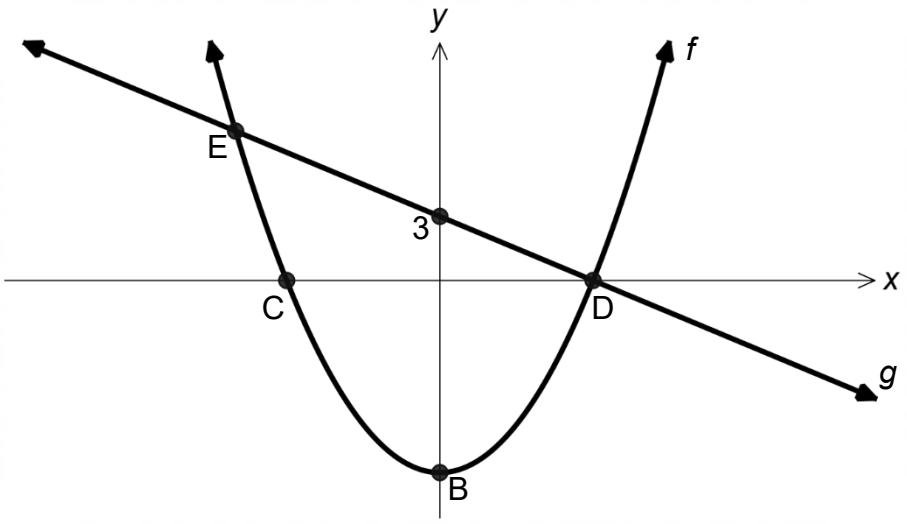
(c) Determine the equation of the graph below. You need not simplify your answer. (4)



[18]

QUESTION 3

The graphs of $f(x) = x^2 - 9$ and $g(x) = mx + c$ are sketched below.



(a) Determine the co-ordinates of C and D. (3)

(b) Write down the co-ordinates of B. (2)

(c) Determine the equation of g. (2)

(d) Determine the co-ordinates of E. (5)

(e) Write down the domain of g. (1)

(f) Write down the range of f. (2)

(g) For which values of x is f increasing? (1)

(h) Use the graph to determine the values of x for which $f(x) \leq 0$. (2)

(i) Determine the minimum value of $y = f(-x) + 5$ (2)

[20]

EXTRA SPACE